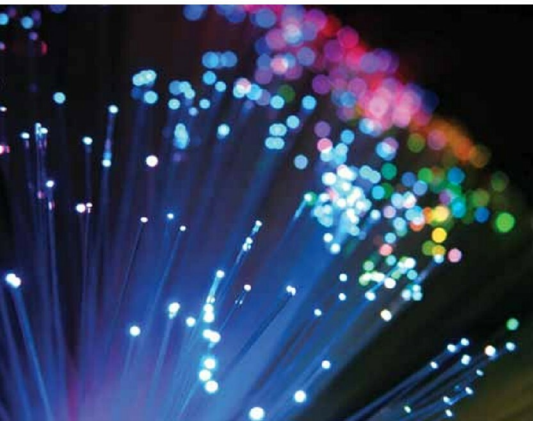


OPTICAL NETWORKS CREATE HUGE DEMAND

Although optical networks have been in existence for many years now, optical communication technologies are continuously evolving. Keeping in sync, optical testing is looking for high-speed, high-performance, all-in-one type and accurate test and measurement solutions that are upgradeable to 40G or 100G in the future



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The need for high-speed connectivity and higher capacity will never cease. The launch of 3G and LTE has fuelled this need for high-speed and high-bandwidth backhaul. To meet this demand, optical networks based on fibre-optics technology provide a stable and upgradable system.

Optical systems have already been dominating the core networks. "Due to their reliability and bandwidth of-

fering, optical networks are making inroads to support the passive optical network technologies in the access networks. Many enterprise networks and residential customers are now connected to core networks via optical networks," shares Mohmedsaeed Mombasawala, general manager-applications, Agilent Technologies.

The increased service needs based on Triple Play in the user segment are fuelling the usage of optical-fibre systems in access networks as well.

There are two kinds of test and measurement requirements for optical communications: Optical network testing during installation and commission of networks, and testing for R&D and manufacturing. From development to deployment, the testing needs are very intensive and knowledge-oriented. Optical testing is looking for high-speed, high-performance, all-in-one type and accurate test and measurement solutions that are upgradeable to 40G or 100G in the future.

Optical communication technologies

An optical communications system is based on the principle of transmission of a signal through an optical fibre to a distant receiver. The electrical signal is converted into optical at the transmitter and back into the original electrical signal at the receiver.

Russel Taws, market manager, JDSU, explains, "The fibre is classified into multi-mode and single-mode based on the way in which the light travels through it. Multi-mode fibres are used to transmit data over relatively short distances (<1 km) and find applications in aerospace, data centres and local area networks. Single-mode fibres can carry signals of 10, 40 and even 100 Gbps over very long distances (>200 km) and are used in all telecommunication systems for voice, data, video and mobile."

"The capacity of a single-mode fibre communication system may be